NAME: _____ Geometry (FL B.E.S.T.)

2.5 Constructing Parallel & Perpendicular

Compass-and-straightedge constructions produce results that are exactly correct, provable through triangle congruence, rather than approximately correct from a measured reading. This section pushes past memorizing steps by asking *why* each construction works — usually because two equal-radius arcs force an SSS congruence, which in turn forces a right angle or equal segment. It also chains the corresponding-angle idea from Section 2.2 into an actual construction for a parallel line, and applies the perpendicular bisector to a genuine problem: locating the center of a circle from three points on it.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Two tools for constructions?

2. Is a protractor allowed in constructions?

3. Construction of $\perp$ at a point on a line uses equal what?

4. This construction creates a line from the point that is what to the given line?

5. Parallels through an external point: how many?

6. Perpendiculars through a point to a line: how many?

7. Perpendicular bisector intersects the segment at its what?

8. Congruent corresponding angles imply lines are what?

9. If an angle formed by two equal-radius arc constructions is $90°$, the lines are what?

10. May you transfer a length by reading a ruler in a construction?

11. Copying an angle relies on which triangle congruence shortcut?

12. Construction through $P \notin \ell$ parallel to ℓ yields a line that never does what with ℓ?

13. Construction of $\perp$ yields what angle measure?

14. A carpenter needs the exact midpoint of a board with no ruler. Which construction should be used, and what two facts does it guarantee simultaneously?

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15. Explain how to construct a 60° angle with no protractor, using only an equilateral triangle built from segment $\overline{AB}$.

✓ Answer Key

1. Compass and straightedge
2. No
3. Equal arcs / equal radii
4. Perpendicular
5. One
6. One
7. Midpoint
8. Parallel
9. Perpendicular
10. No
11. SSS
12. Intersects
13. 90°
14. The perpendicular bisector construction; it guarantees the point is the exact midpoint AND that the bisector line is perpendicular to the board
15. Draw $\overline{AB}$; with radius AB, draw arcs from A and B meeting at C; since $AB = AC = BC$, $\triangle ABC$ is equilateral, so each angle equals 60°